

Topic 5: Arms Race

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March 27, 2026



Overview



Scenario: Purple (P) and Green (G) are two countries competing in an arms race.

Research Question: What happens to the long-term spending of each country?

Key Factors:

- More inclined to increase defence spending as the other's rises
- Less inclined to increase spending as its own spending rises
- Does public sentiment affect defence spending?
- Does limiting spending affect a country's position?

Motivation: It helps predict military escalation between countries, which can inform policies to promote peace and prevent conflict.



Base Model

Model Equations

The equations for the base model are:

$$\frac{\partial p}{\partial t} = ag(t) - bp(t)$$

$$\frac{\partial g}{\partial t} = cp(t) - dg(t)$$

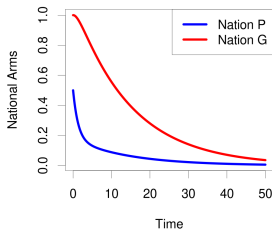
- Feasibility Condition: $(p(t), g(t) \geq 0)$
- a, c : Mutual Fear Factor ($a, c > 0$)
- b, d : Expenditure Factor ($b, d > 0$)

Long Term Behavior (Base Model)



$$bd > ac$$

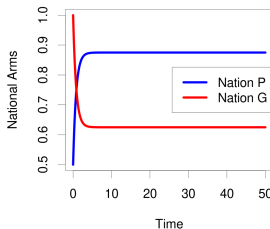
$$a=0.1, b=0.7, c=0.2, d=0.1$$



Complete mutual
disarmament

$$bd = ac$$

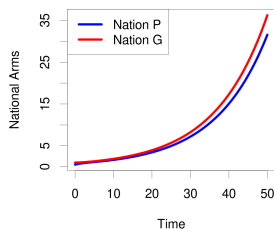
$$a=0.7, b=0.5, c=0.5, d=0.7$$



Stabilized spending

$$bd < ac$$

$$a=0.5, b=0.5, c=0.2, d=0.1$$



Perpetual arms race

Public Sentiment (Trust and Hostility)



Model Equations

The equations after including public sentiment are:

$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r$$
$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s$$

- $(r, s) > 0$ for goodwill and $(r, s) < 0$ for grievance
- Provide further elaborating points

Long Term Behavior (Public Sentiment)



- provide plots
- provide stability conditions and convergences
- This model will take on negative values so discuss that by either capping the value or other methods

Mutual Logistic Fear



Model Equations

The equations after adding logistic growth to account of limited resources are:

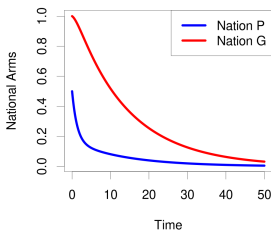
$$\frac{\partial p}{\partial t} = a\left(1 - \frac{p(t)}{K_p}\right)g(t) - bp(t) + r$$

$$\frac{\partial g}{\partial t} = c\left(1 - \frac{g(t)}{K_g}\right)p(t) - dg(t) + s$$

Long Term Behavior (Mutual Logistic Fear)

$$bd > ac$$

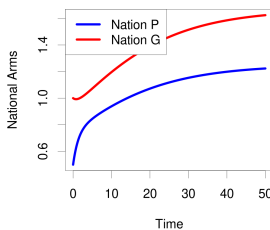
$$a=0.1, b=0.7, c=0.2, d=0.1, K_p=5, K_g=5$$



Complete mutual
disarmament

$$bd = ac$$

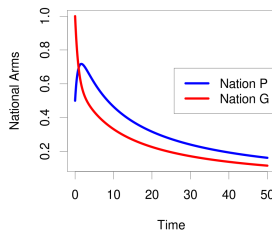
$$a=0.5, b=0.5, c=0.2, d=0.1, K_p=5, K_g=5$$



Stabilized spending

$$bd < ac$$

$$a=0.7, b=0.5, c=0.5, d=0.7, K_p=5, K_g=5$$



Perpetual arms race

Thank you for listening!



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